MA1522 Linear Algebra for Computing

Week 2 Problem Solving Session

Announcements

1. Attempt quiz 1.1 to 1.5, due on Friday 29 August 2025.
2. Tutorial starts next Monday, 8am. Attempt the tutorial questions before coming to class. Find the questions in Canvas files.
3. Make sure you have downloaded MATLAB.

Row Operations in MATLAB

- Input the (augmented) matrix, `R = [1 2 3 4;2 1 1 -1;1 2 -1 0]`. Or define the coefficient matrix and the constant vector separately, `A = [1 2 3;2 1 1;1 2 -1]; b = [4;-1;0];`, then <u>concatenate</u> them together `R = [A b];`.
- To perform $R_2 - 2R_1$, `R(2,:)=R(2,:)-2*R(1,:)`.
- To perform $R_3 \leftrightarrow R_2$, `R([2,3],:)=R([3,2],:)`.
- To perform $-\frac{1}{3}R_3$, `R(3,:)=(-1/3)*R(3,:)`
- Use `format rat` to display values in fractions. Use with caution, it will estimate every number with a fraction, e.g. `pi`.
- Use `format short` for 4 decimal places, and `format long` for 15 decimal places.
- Use `syms a b` for unknowns (symbolic).
- Use `eval` to evaluate. E.g. `a=1; eval 2*a`.

Problem 1

Xiao Ming enjoys eating at vegetable rice stalls. A new stall has just opened in the science canteen, but it does not display the prices of its items. The menu consists of: rice, steamed egg, fried egg, stir-fried kangkong, cabbage, chicken cutlet, and steam fish.

Xiao Ming is curious about the prices of each item. He observes what several customers order and how much they pay:

1. Customer 1 ordered steam egg and cabbage with rice and paid $2.50.
2. Customer 2 ordered fried egg and chicken cutlet with rice and paid $3.0.
3. Customer 3 ordered stir-fried Kangkong, cabbage, and steam fish with rice and paid $4.40.
4. Customer 4 ordered chicken Cutlet and fish with rice and paid $4.30.
5. Customer 5 ordered steamed egg and stir-fried Kangkong with rice and paid $2.70.
6. Customer 6 ordered cabbage and steamed fish with rice and paid $3.50.
7. Customer 7 ordered steamed egg and fried egg with rice and paid $2.50.

Using this information, help Xiao Ming to determine the price of each item.

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x_1 x_2 x_3 x_4 x_5 x_6 x_7

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6. Customer 6 ordered cabbage and steamed fish with rice and paid \$3.50.
7. Customer 7 ordered steamed egg and fried egg with rice and paid \$2.50.

Using this information, help Xiao Ming to determine the price of each item.

Let $x_1, x_2, x_3, x_4, x_5, x_6, x_7$ represent the prices (in dollar) of rice, steamed egg, fried egg, stir-fried kangkong, cabbage, chicken cutlet, and steam fish, respectively. Using the given information, write down a system of linear equations to model the situation.

Problem 1

$$
\begin{cases}
x_1 + x_2 + x_5 = 2.5 \\
x_1 + x_3 + x_6 = 3 \\
x_1 + x_4 + x_5 + x_7 = 4.4 \\
x_1 + x_6 + x_7 = 4.3 \\
x_1 + x_2 + x_4 = 2.7 \\
x_1 + x_5 + x_7 = 3.5 \\
x_1 + x_2 + x_3 = 2.5
\end{cases}
$$

$$
\Rightarrow \begin{cases}
x_1 = 0.8 \\
x_2 = 1 \\
x_3 = 0.7 \\
x_4 = 0.9 \\
x_5 = 0.7 \\
x_6 = 1.6 \\
x_7 = 2
\end{cases}
$$

Problem 1

$$\begin{cases} x_1 + x_2 + x_5 = 2.50 \\ x_1 + x_3 + x_6 = 3.00 \\ x_1 + x_4 + x_5 + x_7 = 4.40 \\ x_1 + x_6 + x_7 = 4.30 \\ x_1 + x_2 + x_4 = 2.70 \\ x_1 + x_5 + x_7 = 3.50 \\ x_1 + x_2 + x_3 = 2.50. \end{cases}$$

$$\left(\begin{array}{ccccccc|c} 1 & 1 & 0 & 0 & 1 & 0 & 0 & 2.5 \\ 1 & 0 & 1 & 0 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 1 & 1 & 0 & 1 & 4.4 \\ 1 & 0 & 0 & 0 & 0 & 1 & 1 & 4.3 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 2.7 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 3.5 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 2.5 \end{array}\right)$$

To key into MATLAB,

```
A = [1 1 0 0 1 0 0 2.5; 1 0 1 0 0 1 0 3; 1 0 0 1 1 0 1 4.4; 1 0 0 0 0 1 1 4.3; 1 1 0
1 0 0 0 2.7; 1 0 0 0 1 0 1 3.5; 1 1 1 0 0 0 0 2.5]
```

Problem 1

Begin row operations:

```
A(2,:)=A(2,:)-A(1,:); A(3,:)=A(3,:)-A(1,:); A(4,:)=A(4,:)-A(1,:);
A(5,:)=A(5,:)-A(1,:); A(6,:)=A(6,:)-A(1,:); A(7,:)=A(7,:)-A(1,:)
A(3,:)=A(3,:)-A(2,:); A(4,:)=A(4,:)-A(2,:); A(6,:)=A(6,:)-A(2,:)
A(4,:)=A(4,:)-A(3,:);A(6,:)=A(6,:)-A(3,:); A(7,:)=A(7,:)+A(3,:)
A(5,:)=A(5,:)+A(4,:); A(6,:)=A(6,:)-A(4,:); A(7,:)=A(7,:)+A(4,:)
A(6,:)=A(6,:)+(1/2)*A(5,:); A(7,:)=A(7,:)-(1/2)*A(5,:)
A(7,:)=A(7,:)-A(6,:)
```

Problem 1

Begin row operations:

```
A(2,:)=A(2,:)-A(1,:); A(3,:)=A(3,:)-A(1,:); A(4,:)=A(4,:)-A(1,:);
A(5,:)=A(5,:)-A(1,:); A(6,:)=A(6,:)-A(1,:); A(7,:)=A(7,:)-A(1,:)
A(3,:)=A(3,:)-A(2,:); A(4,:)=A(4,:)-A(2,:); A(6,:)=A(6,:)-A(2,:)
A(4,:)=A(4,:)-A(3,:);A(6,:)=A(6,:)-A(3,:); A(7,:)=A(7,:)+A(3,:)
A(5,:)=A(5,:)+A(4,:); A(6,:)=A(6,:)-A(4,:); A(7,:)=A(7,:)+A(4,:)
A(6,:)=A(6,:)+(1/2)*A(5,:); A(7,:)=A(7,:)-(1/2)*A(5,:)
A(7,:)=A(7,:)-A(6,:)
```

May reduce to RREF, or use back substitution, get

$$x_1 = 0.8,\ x_2 = 1.0,\ x_3 = 0.7,\ x_4 = 0.9,\ x_5 = 0.7,\ x_6 = 1.5,\ x_7 = 2.0.$$

Problem 1

In this case, we should be using `rref`.

```
A = [1 1 0 0 1 0 0 2.5; 1 0 1 0 0 1 0 3; 1 0 0 1 1 0 1 4.4; 1 0 0 0 0 1 1 4.3; 1 1 0
1 0 0 0 2.7; 1 0 0 0 1 0 1 3.5; 1 1 1 0 0 0 0 2.5];
rref(A)
```

Problem 2

Consider the following augmented matrix of a linear system with variables x_1, x_2, x_3, x_4, x_5.

$$\left(\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{array}\right).$$

Without performing any row operations, determine if the system is consistent. If it is consistent, is the solution unique? If the solution is not unique, which variable should we assign as parameters?

Consistent.

Let $x_3 = s$, $x_5 = t$, $s \in \mathbb{R}$, $t \in \mathbb{R}$

Problem 2

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$$\left(\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{array}\right).$$

Without performing any row operations, determine if the system is consistent. If it is consistent, is the solution unique? If the solution is not unique, which variable should we assign as parameters?

The augmented matrix is in row-echelon form, and since the last column is not a pivot column, the system is consistent. Since there are non-pivot columns in the coefficient side of the augmented matrix, the solution is not unique. We would preferably assign the non-pivot column in the LHS as parameters, that is, $x_3 = s,\ x_5 = t,\ s, t \in \mathbb{R}$.

Problem 2

Consider the following augmented matrix of a linear system with variables x_1, x_2, x_3, x_4, x_5.

$$\left(\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{array}\right).$$

(i) Could we assign x_2 as a parameter? If so, write down a general solution with x_2 as a parameter.

(ii) How about x_1? Write down a general solution with x_1 as a parameter.

(iii) What are the restrictions in assigning parameters in this case?

Problem 2

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$$\left(\begin{array}{ccccc|c} 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 2 \end{array}\right).$$

(i) Could we assign x_2 as a parameter? If so, write down a general solution with x_2 as a parameter.

(ii) How about x_1? Write down a general solution with x_1 as a parameter.

(iii) What are the restrictions in assigning parameters in this case?

There are no restriction on whether a particular variable can be assigned as a parameter. For example, we may assign $x_2 = s$,

$$x_1 = -1 - s - t,\ x_2 = s,\ x_3 = t,\ x_4 = -s,\ x_5 = 2 + s, \quad s, t \in \mathbb{R},$$

or $x_1 = s$,

$$x_1 = s,\ x_2 = -1 - s - t,\ x_3 = t,\ x_4 = 1 + s + t,\ x_5 = 1 - s - t, \quad s, t \in \mathbb{R}.$$

Problem 2

From the equations $x_1 = 1 - x_3 - x_5$, $x_2 = -x_4$ and $x_4 = 2 - x_5$, we conclude that we need 2 parameters, and cannot assign them to the pairs (x_2, x_4), (x_4, x_5), (x_2, x_5).

Questions

1. Is it possible for a system to have exactly 2 solutions?

2. How many parameters are required for a consistent linear system with 5 variables and 3 equations?

3. Can a linear system with n variables has a solution with $\geq n$ parameters?

Questions

1. Is it possible for a system to have exactly 2 solutions?

   An equation with 2 variables is a line, an equation with 3 variables is a plane, etc. Two lines intersect at a point, Two planes intersects along a line, Two equations with 4 variables intersect along a plane, etc. So, the system either has only 1 solution (a point), or the solution contains at least a line.

2. How many parameters are required for a consistent linear system with 5 variables and 3 equations?

   At least 2.

3. Can a linear system with n variables has a solution with $\geq n$ parameters?

   In this case, in the RREF, all columns in the LHS of the augmented matrix must be nonpivot columns, which is only possible if the augmented matrix has all entries 0.

Problem 3

Consider the following system of linear equations in the variables x_1, x_2, x_3, x_4, where a, b are some real numbers:

$$\begin{cases} ax_1 + 2x_2 + ax_3 + (a+b)x_4 = a-b \\ ax_1 + 2x_2 + ax_3 + ax_4 = a-b \\ 3x_1 + 3x_2 - bx_3 + 3x_4 = -b \\ (a+1)x_1 + 3x_2 + (a+1)x_3 + (a+1)x_4 = a-b+1 \end{cases}$$

Solve the system, expressing your answer in terms of a and b where applicable. Discuss any conditions on a and b under which the system has:

(a) a unique solution,

(b) infinitely many solutions,

(c) no solution.

Let's start reducing the system.

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Consider the following system of linear equations in the variables x_1, x_2, x_3, x_4, where a, b are some real numbers:

$$\begin{cases} ax_1 + 2x_2 + ax_3 + (a+b)x_4 = a-b \\ ax_1 + 2x_2 + ax_3 + ax_4 = a-b \\ 3x_1 + 3x_2 - bx_3 + 3x_4 = -b \\ (a+1)x_1 + 3x_2 + (a+1)x_3 + (a+1)x_4 = a-b+1 \end{cases}$$

Solve the system, expressing your answer in terms of a and b where applicable. Discuss any conditions on a and b under which the system has:

(a) a unique solution,

(b) infinitely many solutions,

(c) no solution.

Let's start reducing the system. First leave only 1 nonzero entry in the first column.

```
syms a b;
A = [a 2 a a+b a-b; a 2 a a a-b; 3 3 -b 3 -b; a+1 3 a+1 a+1 a-b+1]
A(2,:)=A(2,:)-A(1,:); A(4,:)=A(4,:)-A(1,:)
A(3,:)=A(3,:)-3*A(4,:); A(1,:)=A(1,:)-a*A(4,:)
A([1,4],:)=A([4,1],:)
```

Problem 3

$$\left(\begin{array}{cccc|c} a & 2 & a & a+b & a-b \\ a & 2 & a & a & a-b \\ 3 & 3 & -b & 3 & -b \\ a+1 & 3 & a+1 & a+1 & a-b+1 \end{array}\right) \longrightarrow \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1-b & 1 \\ 0 & 2-a & 0 & b(a+1) & -b \\ 0 & 0 & -b-3 & 3b & -b-3 \\ 0 & 0 & 0 & -b & 0 \end{array}\right).$$

Problem 3

$$\left(\begin{array}{cccc|c} a & 2 & a & a+b & a-b \\ a & 2 & a & a & a-b \\ 3 & 3 & -b & 3 & -b \\ a+1 & 3 & a+1 & a+1 & a-b+1 \end{array}\right) \longrightarrow \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1-b & 1 \\ 0 & 2-a & 0 & b(a+1) & -b \\ 0 & 0 & -b-3 & 3b & -b-3 \\ 0 & 0 & 0 & -b & 0 \end{array}\right).$$

```
A(2,:)=A(2,:)+(a+1)*A(4,:); A(3,:)=A(3,:)+3*A(4,:); A(1,:)=A(1,:)-A(4,:)
simplify(A)
```

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2-a & 0 & 0 & -b \\ 0 & 0 & -b-3 & 0 & -b-3 \\ 0 & 0 & 0 & -b & 0 \end{array}\right).$$

Problem 3

$$\left(\begin{array}{cccc|c} a & 2 & a & a+b & a-b \\ a & 2 & a & a & a-b \\ 3 & 3 & -b & 3 & -b \\ a+1 & 3 & a+1 & a+1 & a-b+1 \end{array}\right) \longrightarrow \left(\begin{array}{cccc|c} 1 & 1 & 1 & 1-b & 1 \\ 0 & 2-a & 0 & b(a+1) & -b \\ 0 & 0 & -b-3 & 3b & -b-3 \\ 0 & 0 & 0 & -b & 0 \end{array}\right).$$

```
A(2,:)=A(2,:)+(a+1)*A(4,:); A(3,:)=A(3,:)+3*A(4,:); A(1,:)=A(1,:)-A(4,:)
simplify(A)
```

$$\left(\begin{array}{cccc|c} 1 & 1 & 1 & 1 & 1 \\ 0 & 2-a & 0 & 0 & -b \\ 0 & 0 & -b-3 & 0 & -b-3 \\ 0 & 0 & 0 & -b & 0 \end{array}\right).$$

Observe from second row that if $a = 2$ and $b \neq 0$, then system has no solution.

Problem 3

Suppose $a = 2$ and $b = 0$, then system has infinitely many solutions with 2 parameters:

```
a=2;b=0; rref(subs(A))
```

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

General solution:

$$x_1 = -s - t, \quad x_2 = s, \quad x_3 = 1, \quad x_4 = t, \quad s, t \in \mathbb{R}.$$

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```
a=2;b=0; rref(subs(A))
```

$$\left(\begin{array}{cccc|c} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

General solution:

$$x_1 = -s - t, \quad x_2 = s, \quad x_3 = 1, \quad x_4 = t, \quad s, t \in \mathbb{R}.$$

Now assume throughout that $a \neq 2$. If $b = 0$, system has infinitely many solutions with 1 parameter:

```
syms a; b=0; rref(subs(A))
```

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

General solution:

$$x_1 = -s, \quad x_2 = 0, \quad x_3 = 1, \quad x_4 = s, \quad s \in \mathbb{R}.$$

Problem 3

If $b = -3$, the system has infinitely many solutions with 1 parameter:

```
b=-3;rref(subs(A))
```

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & (a+1)/(a-2) \\ 0 & 1 & 0 & 0 & -3/(a-2) \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

General solution:

$$x_1 = \frac{a+1}{a-2}, \quad x_2 = -\frac{3}{a-2}, \quad x_3 = s, \quad x_4 = 0, \quad s \in \mathbb{R}.$$

Problem 3

If $b = -3$, the system has infinitely many solutions with 1 parameter:

```
b=-3;rref(subs(A))
```

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & (a+1)/(a-2) \\ 0 & 1 & 0 & 0 & -3/(a-2) \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

General solution:

$$x_1 = \frac{a+1}{a-2}, \quad x_2 = -\frac{3}{a-2}, \quad x_3 = s, \quad x_4 = 0, \quad s \in \mathbb{R}.$$

Finally, if $b \neq -3, 0$, then the system has a unique solution:

```
syms a b; rref(A)
```

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 0 & -b/(a-2) \\ 0 & 1 & 0 & 0 & b/(a-2) \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{array}\right).$$

Unique solution:

$$x_1 = -\frac{b}{a-2}, \quad x_2 = \frac{b}{a-2}, \quad x_3 = 1, \quad x_4 = 0.$$

Problem 4

Write down a linear system with 5 variables x_1, x_2, x_3, x_4, x_5 and 3 equations whose general solution is

$$x_1 = s - 2t - 1,\ x_2 = s,\ x_3 = 3 + 3t,\ x_4 = 1 + t,\ x_5 = t, \quad s, t \in \mathbb{R}.$$

Problem 4

Write down a linear system with 5 variables x_1, x_2, x_3, x_4, x_5 and 3 equations whose general solution is

$$x_1 = s - 2t - 1,\ x_2 = s,\ x_3 = 3 + 3t,\ x_4 = 1 + t,\ x_5 = t, \quad s, t \in \mathbb{R}.$$

The linear system does not contain any parameters.

Problem 4

Write down a linear system with 5 variables x_1, x_2, x_3, x_4, x_5 and 3 equations whose general solution is

$$x_1 = s - 2t - 1,\ x_2 = s,\ x_3 = 3 + 3t,\ x_4 = 1 + t,\ x_5 = t, \quad s, t \in \mathbb{R}.$$

The linear system does not contain any parameters. Substitute $s = x_3$ and $t = x_5$ to the other identities, we get

$$x_1 = x_2 - 2x_4 - 1, \quad x_3 = 3 + 3x_5, \quad x_4 = 1 + x_5.$$

Problem 4

Write down a linear system with 5 variables x_1, x_2, x_3, x_4, x_5 and 3 equations whose general solution is

$$x_1 = s - 2t - 1,\ x_2 = s,\ x_3 = 3 + 3t,\ x_4 = 1 + t,\ x_5 = t, \quad s, t \in \mathbb{R}.$$

The linear system does not contain any parameters. Substitute $s = x_3$ and $t = x_5$ to the other identities, we get

$$x_1 = x_2 - 2x_4 - 1, \quad x_3 = 3 + 3x_5, \quad x_4 = 1 + x_5.$$

Convert to standard form, a corresponding linear system is thus

$$\begin{cases} x_1 - x_2 + 2x_4 = -1 \\ x_3 - 3x_5 = 3 \\ x_4 - x_5 = 1 \end{cases}$$

Challenge

For a fixed number of variables and equations, is there a one-to-one correspondence between the set of solution sets of linear systems and the set of reduced row-echelon forms of their augmented matrices?

Submission

- Only need to submit your answers for those labeled as Problem X.
- If you have written your answer on an A4 paper, to submit, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file (make sure they are in order of the page).
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by **StudentNo_week2** (e.g. **A1234567Z_week2.pdf**).
- Submit the PDF file to Canvas assignments, Lecture group 1: Problem Solving Session Week 2.
- Deadline: Today 2359hrs.